EARTHPRESS · PROBLEM STATEMENT

EarthPress problem statement

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-25	Amish Chadha	Kit 1.8 users-and-needs and stakeholders tables (Tables 1 and 2), the source of EPR-URS-001; press mass from the kit 1.9.1 model (203.8 kg)
0.5	2026-10-02	Amish Chadha	Partner rule and first candidate; design soil and lime option; press mass (EPR-DEC-001, 2026-10-02)
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Open questions updated after EPR-CAL-001 and EPR-DDR-001 (block size and stabilizer adopted for TRL 3; two operators needed on paper)
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (users, context, constraints, prior work with sources, open questions)
0.1	2026-09-25	Amish Chadha	Initial scaffold

EarthPress problem statement

Walls made of fired brick or cement block carry high embodied carbon and cost, and the cheaper, lower-carbon alternative, compressed stabilized earth blocks (CSEB), depends on a press that most small builders cannot buy or make. Open press designs exist, but the best documented one is hydraulic and costs thousands of dollars, while the manual lever presses that suit a village building site are sold as closed products or survive as old, undimensioned drawings. EarthPress aims to fill that gap with an open, welder-buildable manual press and a field soil test kit.

Why it matters

- The buildings and construction sector accounts for about 32 % of global energy use and 34 % of global CO2 emissions (UNEP, *Global Status Report for Buildings and Construction 2024/2025*).
- Cement production alone accounts for around 8 % of global CO2 emissions (Chatham House, *Making Concrete Change*, 2018).
- UN-Habitat estimates that by 2030 about 3 billion people, about 40 % of the world's population, will need access to adequate housing, or about 96,000 new affordable units every day (UN-Habitat, Housing).
- 1.12 billion people lived in slums or informal settlements in 2022, 130 million more than in 2015 (UN, *Sustainable Development Goals Report 2024*, Goal 11).
- The Auroville Earth Institute reports that CSEB stabilized with 5 % cement embody about 548 MJ/m³ and about 49 kg CO2/m³, against about 6,120 MJ/m³ and about 643 kg CO2/m³ for locally fired bricks, that is, about 11 times less energy and 13 times less pollution (Auroville Earth Institute, CSEB).

Users and context

EarthPress serves people who build small load-bearing or infill walls with the soil on or near their site.

User	Need	Context
Self-builders and families	Make their own walling for a one- or two-room house, a store or an extension	Peri-urban and rural sites with clay-sand soil; little cash, plenty of labor
Small block-making enterprises	Sell blocks locally as a business	A yard with shade, water and a stock of soil; a crew of three to five
NGOs and community builders	Train masons and build schools, clinics or reconstruction housing	Programs in sub-Saharan Africa, South Asia and Latin America
Local welders and fabricators	Build, repair and sell the press	A stick welder, angle grinder and drill press; steel from a local stockist
Architects and owner-builders in high-income countries	Low-carbon garden walls, studios and small buildings	Places with earthen building codes, such as New Mexico

The press works outdoors on uneven ground, in dust, heat and rain, and is moved between sites on a pickup or by hand. Operators range from trained masons to first-time volunteers.

Users and needs

Table 1. Users and needs (the source of the user requirements in EPR-URS-001).

User type	Goals and needs	Typical use scenario	Pain points and challenges
Self-builders and families	Make walling for a one- or two-room house from site soil at low cash cost; blocks of a size local masons lay	A family and two helpers press for a week or two beside the house plot	Little cash; no power on site; first-time operators; soil of unknown quality; heavy lifting
Masons and builders	Uniform, square blocks of one height so walls run true; enough blocks a day to keep a crew laying	A mason checks blocks with the gauge and lays 8 to 9 m2 of wall a day	Blocks that vary in height or crumble at the edges; slow supply
Small block-making enterprises	Steady output (about 300 blocks a day), low running cost, a press that lasts and can be repaired	A crew of four runs a yard in shade with a soil heap and a water drum	Wear of mold and pins; downtime waiting for parts; cement cost
NGO and community trainers	A press and soil tests that volunteers learn in a day, safe enough for mixed crews, documented for many sites	A program trains masons and builds schools or reconstruction housing	Injuries from levers and pinch points; unclear soil suitability; equipment that cannot be moved between villages
Local welders and fabricators	Drawings and a parts list for common sections, with no machining beyond drilling and bought pins	A small shop builds a press in a few days from stockist steel and sells or repairs it	Undimensioned or closed designs; parts that need a lathe; unclear assembly order
Architects and owner-builders in high-income countries	Blocks that meet an earthen building code; lower embodied carbon	A studio or garden wall where earthen codes allow CEB	Proving block strength to an inspector; finding a press

Stakeholders

Table 2. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Co-design partner (first candidate: Auroville Earth Institute, decided 2026-10-02)	Trains builders in CSEB; local soils, cement supply and block testing	Co-design sessions, soil compaction tests, review of the requirements and the build plan
Building authorities and code bodies (for example New Mexico, Bureau of Indian Standards)	Set block strength and testing rules	Check the applicable standard per region (EPR-REG-001) before any structural use
Funders and humanitarian programs	Pay for housing and training; need safe, durable outcomes	Safety case, cost per house, training plan
Cement and lime suppliers	Supply the stabilizer	Local price and availability for the mix chart
Steel stockists and machine shops	Supply sections, plate and turned pins	Local sizes and prices; 42CrMo4 bar and 40 x 46 mm bushes (EPR-DEC-001, to confirm)
Testing laboratories	Test block strength and durability	Block tests at TRL 4 (on hold)

Constraints

- Garage-buildable prototype for about $500 USD in parts (`project.yaml`), with no machining beyond drilling and purchased turned pins.
- Human powered: no electricity, hydraulics or engine on site.
- Parts from local steel stockists: flat bar, plate, channel, rectangular tube and pipe in common sizes.
- Wear parts and pins replaceable with hand tools.
- Movable by four people, or by two people once dismantled (EPR-CAL-001: about 204 kg in 11 pieces of 45 kg or less).
- Blocks must meet a recognized strength reference for the intended use (for example Auroville class B, or the New Mexico code minimum of 300 psi, about 2.1 MPa) when made from suitable, tested soil.

Out of scope

- Hydraulic or motorized presses, and block-making plant.
- Structural design of buildings, including seismic design; EarthPress supplies blocks, not building approval.
- Rammed earth, adobe and fired brick.
- Certification of blocks against any national standard (a later-stage activity with a test laboratory).

Prior work

Work	What it offers	Gap for EarthPress
CINVA-Ram, a manual lever press designed by Raúl Ramírez at CINVA, Bogotá, in 1956 and patented in 1958 (Botti, 2023)	The original single-lever, bottom-piston press; widely copied	Low compaction by modern standards; drawings vary and are rarely open-licensed
Auroville Earth Institute and its Auram presses (India)	Mature CSEB practice: 5 % cement on average, compaction pressure of 2 to 4 MPa, strength classes; the Auram 3000 makes about 1,000 blocks a day (Auroville Earth Institute)	Commercial presses; not open hardware
Open Source Ecology CEB Press	Open hardware (CC BY-SA 4.0, GPLv3), about 6 blocks a minute; materials cost $3,000 to $6,500, about $10,000 assembled (OSE wiki)	Hydraulic and engine powered; too costly for a self-builder
2021 New Mexico Earthen Building Materials Code, 14.7.4 NMAC	Covers CEB; cured units need 300 psi compressive strength and 50 psi modulus of rupture, with five random units tested per project (NM Commission of Public Records)	A strength target, not a press design
Interlocking stabilized soil block (ISSB) presses used in East Africa	Mortar-saving interlocking blocks	Specific block geometries; mostly commercial presses

User research and co-design

This design is for communities the author is not part of, so requirements come from the people who will use it.

- [ ] Identify a local partner organization. Decided on 2026-10-02 (EPR-DEC-001): a partner that already trains builders in stabilized earth blocks and has local soils and cement supply; first candidate to approach, the Auroville Earth Institute in India. Nothing is agreed yet.
- [ ] Run co-design sessions with intended users; record who, where and what was learned
- [ ] Validate load, distance, terrain and cost assumptions in the field
- [ ] Revise the user and product requirements (EPR-URS-001, EPR-PRS-001) from findings before freezing the design

Open questions

- The flat 290 x 140 x 90 mm block was decided by Amish on 2026-09-25 (EPR-DDR-001 item 3, EPR-DDR-002). Do local masons prefer it, a larger module or an interlocking block?
- EPR-CAL-001 finds that one operator pulling 500 N reaches only about 0.5 MPa and that two people are needed on the lever for 2 MPa. Amish decided on 2026-09-25 to go with a two-person T-handle (EPR-DDR-002); co-design should still confirm that builders accept it rather than a lower pressure and more cement.
- Which soils in the first partner region pass the field tests, what cement or lime content do they need, and how stiff are they in the press? Until they are measured, the operating chart uses the softer soil (809 N, 405 N per operator), and lime is a documented option with its own chart and longer curing (decided 2026-10-02).
- What block strength do local building rules or funders require, and who will test blocks?
- How far do blocks and the press travel, and on what vehicle? The press is about 204 kg with its guard, bolts, bushes and weld metal, in pieces of 45 kg or less, against the 200 kg R7 total set on 2026-10-02 (decision D2 in EPR-DEC-001, proposed, awaiting Amish).